

Abdoul Malick Oumarou Allassane

Software Developer

Sherbrooke, Québec, Canada · 819 861-2635 · abdoulmalickoumarouallassane@gmail.com

linkedin.com/in/oumarou-allassane-abdoul-malick-46b2a5259 · github.com/OUMAROU-ALLASSANE · oumarou-allassane.github.io

PROFILE

Final-year Computer Science and Image Science student at Université de Sherbrooke, graduating December 2026. Three years of continuous employment with the same company, two of them as team lead responsible for code, electronics and the supervision of 6 to 12 people. I design and ship complete applications: MVC architecture, React, Python and Django, PostgreSQL, Docker, automated testing and deployment. Team lead on the final-year capstone project, a web-based project management platform built by a four-person team using Scrum. Available full time from January 2027.

TECHNICAL SKILLS

Programming languages: Python, C, C++, C#, Java, JavaScript, SQL, Julia

Web development and databases: React, Django, REST APIs, PostgreSQL, HTML, CSS, entity-relationship modelling, normalisation

Tools and methods: Git, GitHub, Docker, Linux, Unix, macOS, Makefile, pytest, unit testing, code review, Scrum, agile methods, UML, design patterns, MVC architecture

Machine learning and data: PyTorch, diffusers, ControlNet, Stable Diffusion, supervised learning, SVM and kernel methods, Bayesian models, Gaussian mixtures, EM algorithm, dimensionality reduction, clustering, regression, hypothesis testing, nonlinear optimisation

Computer vision and image processing: OpenCV, MediaPipe, NumPy, Pillow, filtering and convolution, edge detection, mathematical morphology, stereo vision, camera calibration, 3D reconstruction, Fourier transform, DFT, FFT, DCT, wavelets, Radon transform, RGB and CIELAB colour spaces

Embedded systems and networking: ESP32, Arduino, Raspberry Pi, C and C++ firmware, microcontrollers, sensors, LED animation, electronic wiring, IPv4 protocol, concurrent programming, process synchronisation

Computer graphics and games: Unity, Unreal Engine, graphics pipeline, shaders, GPU programming, meshes, illumination models, path tracing, Monte-Carlo methods, FFmpeg

PROFESSIONAL EXPERIENCE

Team Lead — Technical and Operations Manager

August 2024 – present

Fox in a Box: Jeux d'évasion inc. — Sherbrooke, Quebec

- Supervise a team of 6 to 12 game masters: scheduling, onboarding new staff and allocating work on the floor.
- Hold technical ownership of every escape room: code, electronics and puzzles, in commercial operation seven days a week.
- Carry out preventive and corrective maintenance, diagnosing and repairing failures between two customer groups with no interruption of service.
- Manage inventory and procurement: supplier orders and purchasing, with the company credit card under my responsibility.

Intern — Embedded Software Developer (T-1 co-op internship)

Summer 2024

Fox in a Box: Jeux d'évasion inc. — Sherbrooke, Quebec

- Built the automation of the Pirate escape room within a three-intern team supervised by the company owner.
- Programmed C and C++ firmware on ESP32 and Arduino microcontrollers, and central orchestration on Raspberry Pi.
- Implemented IPv4 network communication between the room's modules.
- Designed LED animations and carried out the full electronic wiring, from design to commissioning.
- Independently learned Raspberry Pi programming and its interfacing with existing systems despite limited internal documentation; the project was delivered and the room has been in operation since.

Game Master

November 2023 – summer 2024

Fox in a Box: Jeux d'évasion inc. — Sherbrooke, Quebec

- Ran games, welcomed customers, handled bookings and payments, and provided bistro service.

Bar Back

2022 – 2023

Splendido, Big Mamma group — Marseille, France

- Independently handled setup, restocking and stock rotation for the bar of a restaurant with more than 500 seats.

PROJECTS

Collaborative Project Management Platform — Team Lead

IFT596, 6-credit capstone project, 4-person team · 2026

Technologies: React, Python, Django, PostgreSQL, Docker, MVC architecture, Scrum, GitHub Spec Kit

- Led a four-person team with varied backgrounds: work allocation, code reviews and progress tracking.
- Designed a web application centralising tasks and events: Kanban board, roles and permissions, notifications, change history and a progress dashboard.
- Produced the engineering deliverables: functional and non-functional requirements, architecture, work breakdown structure, test strategies and deployment plan.
- Designed and owned the points-and-levels gamification system, along with the interface and user experience work.

animpipe — AI Animation Production Pipeline

Personal project, designed and built solo · 2026

Technologies: Python, PyTorch on MPS backend, Stable Diffusion, ControlNet, diffusers, FFmpeg, OpenCV, pydantic, pytest, Makefile

- Designed an eight-stage pipeline turning an iPhone video into a stylised animation, running locally on an Apple Silicon GPU.
- Structured every stage as an independent, testable command-line tool orchestrated by a Makefile.
- Guaranteed reproducibility through validated YAML configuration, a fixed seed and a JSON manifest recording parameters, model versions and file hashes.
- Implemented a hash-based cache allowing an interrupted run to resume without recomputation; roughly 2,400 lines of Python covered by a pytest suite.

Real-time Audio Visualiser

Personal project · 2026

Technologies: Python, Fourier transform, spectral analysis, signal processing

- Renders two switchable live representations of a track: time-domain waveform and a frequency spectrum computed by Fourier transform.
- Applies real-time audio effects during playback: echo, low-pass filter, bitcrush and reverb, with variable playback speed.
- Loads a local file or searches for a track online, and displays synchronised .lrc lyrics.

EDUCATION

Bachelor of Science, Computer Science and Image Science

2023 – 2026

Université de Sherbrooke — Sherbrooke, Quebec

Co-op program. Graduating December 2026. All 69 core program credits completed.

Third year, Bachelor of Mathematics

2022 – 2023

Aix-Marseille Université — Marseille, France

Undergraduate Diploma, Mathematics and Physics

2020 – 2022

Université privée de Sousse — Sousse, Tunisia

Relevant Coursework: Object-Oriented Design Methods (IFT232) · Data Structures (IFT339) · Algorithms and Data Structures (IFT436) · Database Fundamentals (IFT187) · Concurrent Processes and Parallelism (IFT630) · Development Techniques and Tools (IGL201) · Major Software Project (IFT596) · Interfaces and Multimedia (IFT215) · Data Science (IFT599) · Machine Learning Techniques (IFT603) · Introduction to Cyberattacks (IFT508) · Image Analysis (IMN259) · 3D Vision (IMN269)

AWARDS AND SCHOLARSHIPS

Republic of Niger Excellence Scholarship, awarded to the top graduate of the 2020 science class · Academic Success and Excellence Scholarship, Université de Sherbrooke, winter 2024 · Tuition Supplement Exemption, Government of Quebec

Languages: French — working and academic language. English — intermediate level, fluent reading of technical documentation.